



University of Central Punjab
FOIT&CS
Final Term Exam

Course Title: Linear Algebra - (All Sections)

Course Coordinator: Dr. Maria Naseem

Course Code: CSSS2743

Marks: 60

Time: 2.5 hr.

Semester: Fall 23

Name:

Registration Number:

INSTRUCTIONS

1. Write your name and registration number on the Question Paper and Answer Sheet.
 2. Write with blue/black permanent ink pen.
 3. All your rough work and calculations should also be available on the answer sheet.
 4. Make sure your calculator is in radian mode. Exchange of calculators is not allowed.
- No cheat sheet, notes, handbooks or any kind of sharing allowed.

ATTACH THE QUESTION PAPER WITH ANSWER SHEET.

Q1. [Marks: 4 + 8 = 12]

In a manufacturing plant, three types of products A, B, and C are produced. The plant uses three machines M1, M2, and M3 to produce these products. Each machine has different efficiencies in producing each product. The production rates (in units per hour) for each machine and product are given in the following matrix:

$$M = \begin{bmatrix} 3 & 2 & 4 \\ 2 & 4 & 3 \\ 4 & 3 & 2 \end{bmatrix} \quad P = \begin{bmatrix} 106 \\ 107 \\ 102 \end{bmatrix}$$

where the rows represent machines (M1, M2, M3) and the columns represent products (A, B, C). If the plant aims to produce 106 units of product A, 107 units of product B, and 102 units of product C per hour, determine the number of hours each machine needs to operate to meet the production targets.

- a. Set up a system of linear equations for the given scenario.
- b. Solve the linear system by applying Gauss Jordan Method to determine the required answers.

Q2. [Marks: 6 + 6 = 12]

- a. Consider the following equation of circle

$$(x - 2)^2 + (y - 3)^2 = 4$$

Find its reflection along the line $y = -x$. Sketch the original and reflected circle.

- b. Find the equation of unit square after scaling it by factor 3 and then translate the scaled square by the vector (1, 2). Sketch the final image.

$$\frac{-109}{3}$$

✓ Q3. [Marks: 8 + 2 = 10]

✓ a. Encode the following using the key matrix $K = \begin{bmatrix} 3 & 1 \\ 3 & 4 \end{bmatrix}$.

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✓ b. Find the inverse of K under modulo 26.

Q4. [Marks: 4 + 4 + 4 + 2 = 14]

Consider the following vectors from $\mathbb{R}^3$.

$$S = \{A = (1,2,3), B = (1,1,2), C = (2,0,2)\}$$

- ✓ a. Determine whether the vector $D = (3,4,3)$ can be expressed as a linear combination of A, B , and C .
- ✓ b. Determine whether S is linearly independent.
- c. Determine whether the set S spans the $\mathbb{R}^3$ space. Justify your answer.
- d. Can S be basis for the $\mathbb{R}^3$ space. Justify your answer

Q5. [Marks: 6 + 1 + 5 = 12]

Imagine a scenario where you have a social network represented by a 3×3 matrix A where each entry a_{ij} represents the strength of the relationship between individuals i and j . For example, $a_{ij} = 2$ means that individuals i and j have a strong connection. Here's the matrix representing this social network:

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 3 \end{bmatrix}$$

- ✓ a. Compute the eigenvalues of matrix A .
- ✓ b. Interpret the meaning of these eigenvalues in the context of the social network.
- ✓ c. Find the corresponding eigenvector for any one of the eigenvalues.

Happy Exam Season! ☺